The Nexus Triple Product Fusion Model: A Nonlinear Dynamical, Control, and Verification-Oriented Specification

Mohammad Saad Younus

Abstract

We formalize the Nexus Fusion Model as a nonlinear controlled dynamical system with a triple-product threshold, positivity and well-posedness properties, equilibrium structure, local linear and nonlinear stability, a finite-horizon optimal-control formulation, and a verification-oriented finite-state abstraction. We also state sufficient conditions for forward invariance of Fusion Mode, reachability under bounded controls, completeness of the abstraction under transversal guard crossings, and global ultimate boundedness with an input-to-state style estimate.

1 Introduction

The Nexus Fusion Model is a three-state nonlinear control system with threshold-defined operating modes. Its core performance functional is the triple product

$$\Phi(x) = EN\tau.$$

The framework combines nonlinear dynamics, Lyapunov stability, optimal control, and hybrid abstraction in a single mathematical specification.

2 State Space and Variables

Let $t \in \mathbb{R}_{\geq 0}$. Define

$$x(t) = \begin{pmatrix} E(t) \\ N(t) \\ \tau(t) \end{pmatrix} \in \mathbb{R}^3_{\geq 0},$$

where:

- $E(t)$ is the lattice energy,
- $N(t)$ is the active node density,
- $\tau(t)$ is the stability window.

Assume

$$E, N, \tau \in C^1([0, \infty)).$$

3 Fusion Functional

Definition 1 (Nexus Triple Product)**.** *The Nexus Triple Product is*

$$\Phi(x) := EN\tau.$$

Definition 2 (Fusion Mode)**.** *Let* $C_{\text{Nexus}} > 0$*. The system is in Fusion Mode at time* t *if and only if*

$$\Phi(x(t)) \geq C_{\text{Nexus}}.$$

4 System Architecture Diagram

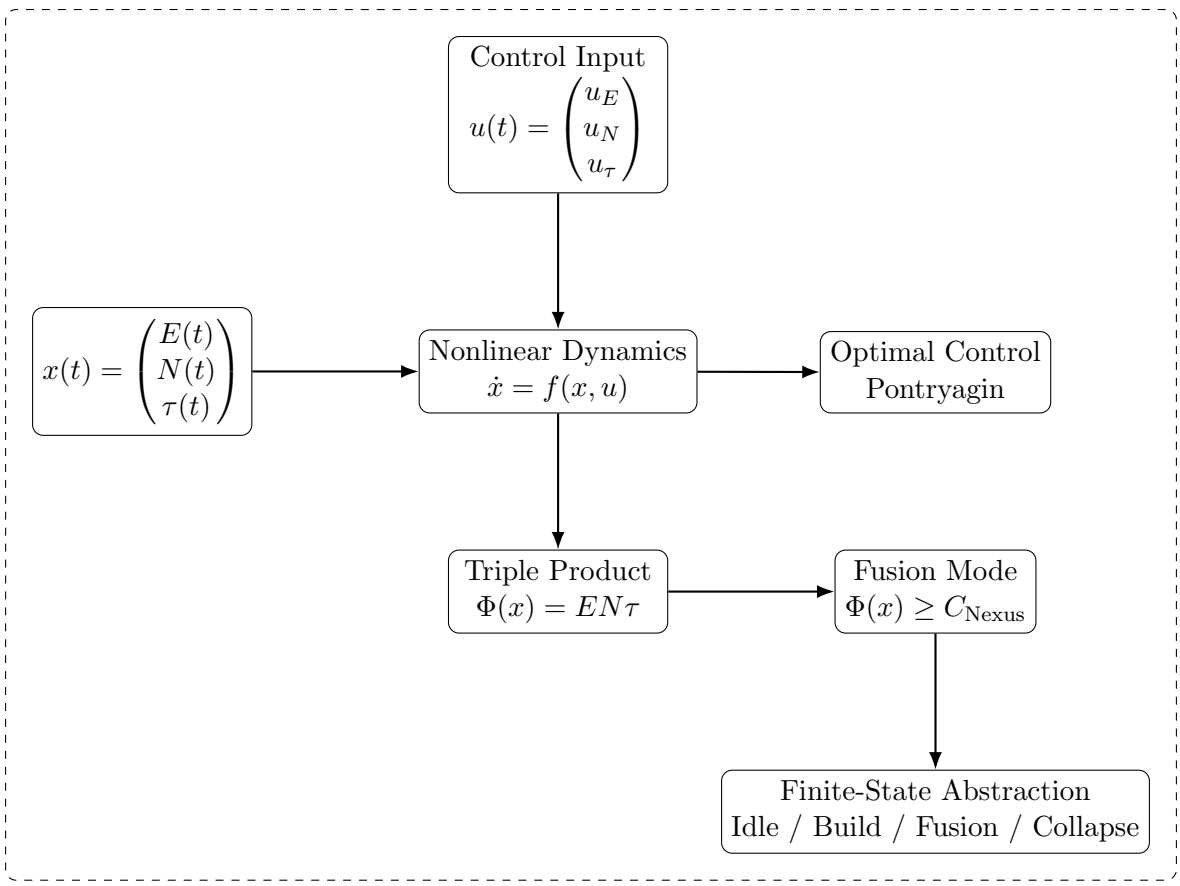

Figure 1: Architecture of the Nexus Fusion Model.

5 Controlled Nonlinear Dynamics

Let

$$u(t) = \begin{pmatrix} u_E(t) \\ u_N(t) \\ u_\tau(t) \end{pmatrix} \in \mathbb{R}^3_{\geq 0}$$

denote the control input.

The system evolves according to

$$\dot{E} = -\beta_E E + \alpha_E u_E + \kappa_{EN} EN + \kappa_{E\tau} E\tau - \gamma_E E^2, \tag{1}$$
$$\dot{N} = -\beta_N N + \alpha_N u_N + \kappa_{NE} EN + \kappa_{N\tau} N\tau - \gamma_N N^2, \tag{2}$$
$$\dot{\tau} = -\beta_\tau \tau + \alpha_\tau u_\tau + \kappa_{\tau E} E\tau + \kappa_{\tau N} N\tau - \gamma_\tau \tau^2. \tag{3}$$

Here:

- $\beta_E, \beta_N, \beta_\tau > 0$ are decay coefficients,
- $\alpha_E, \alpha_N, \alpha_\tau > 0$ are control gains,
- $\kappa_{EN}, \kappa_{E\tau}, \kappa_{NE}, \kappa_{N\tau}, \kappa_{\tau E}, \kappa_{\tau N} \geq 0$ are nonlinear coupling coefficients,
- $\gamma_E, \gamma_N, \gamma_\tau > 0$ are saturation coefficients.

6 Well-Posedness and Positivity

Assumption 1. *The control input satisfies*

$$u \in L^\infty_{\text{loc}}([0,\infty); \mathbb{R}^3_{\geq 0}).$$

Proposition 1 (Local existence and uniqueness)**.** *For every initial condition $x(0) = x_0 \in \mathbb{R}^3_{\geq 0}$ and every locally essentially bounded control u, the system admits a unique maximal solution.*

Proposition 2 (Positive invariance)**.** *If $x(0) \in \mathbb{R}^3_{\geq 0}$ and $u(t) \in \mathbb{R}^3_{\geq 0}$ for all $t \geq 0$, then*

$$x(t) \in \mathbb{R}^3_{\geq 0} \quad \textit{for all times in the interval of existence.}$$

Proof. On the coordinate boundaries,

$$\dot{E}|_{E=0} = \alpha_E u_E \geq 0, \qquad \dot{N}|_{N=0} = \alpha_N u_N \geq 0, \qquad \dot{\tau}|_{\tau=0} = \alpha_\tau u_\tau \geq 0.$$

Hence trajectories cannot leave the nonnegative orthant. □

7 Equilibrium Analysis

Assume constant controls

$$u(t) \equiv \bar{u} = \begin{pmatrix} \bar{u}_E \\ \bar{u}_N \\ \bar{u}_\tau \end{pmatrix}.$$

An equilibrium $x^* = (E^*, N^*, \tau^*)^\top$ satisfies

$$0 = -\beta_E E^* + \alpha_E \bar{u}_E + \kappa_{EN} E^* N^* + \kappa_{E\tau} E^* \tau^* - \gamma_E (E^*)^2, \tag{4}$$
$$0 = -\beta_N N^* + \alpha_N \bar{u}_N + \kappa_{NE} E^* N^* + \kappa_{N\tau} N^* \tau^* - \gamma_N (N^*)^2, \tag{5}$$
$$0 = -\beta_\tau \tau^* + \alpha_\tau \bar{u}_\tau + \kappa_{\tau E} E^* \tau^* + \kappa_{\tau N} N^* \tau^* - \gamma_\tau (\tau^*)^2. \tag{6}$$

Proposition 3 (Steady-state fusion)**.** *If there exists an equilibrium $x^* \in \mathbb{R}^3_{>0}$ such that*

$$\Phi^* := E^* N^* \tau^* \geq C_{\text{Nexus}},$$

then the system admits a steady-state fusion regime.

8 Linearized Stability

Let $f(x, u)$ denote the right-hand side of the state equations and define

$$J(x, u) := \frac{\partial f}{\partial x}(x, u).$$

At an equilibrium $(x^*, \bar{u})$, the Jacobian is

$$J^* = \begin{pmatrix} -\beta_E + \kappa_{EN}N^* + \kappa_{E\tau}\tau^* - 2\gamma_E E^* & \kappa_{EN}E^* & \kappa_{E\tau}E^* \\ \kappa_{NE}N^* & -\beta_N + \kappa_{NE}E^* + \kappa_{N\tau}\tau^* - 2\gamma_N N^* & \kappa_{N\tau}N^* \\ \kappa_{\tau E}\tau^* & \kappa_{\tau N}\tau^* & -\beta_\tau + \kappa_{\tau E}E^* + \kappa_{\tau N}N^* - 2\gamma_\tau \tau^* \end{pmatrix}.$$

Theorem 1 (Local asymptotic stability via linearization)**.** *If all eigenvalues of J^* have strictly negative real parts, then the equilibrium x^* is locally asymptotically stable.*

9 Nonlinear Stability

Let $x^* = (E^*, N^*, \tau^*)^\top$ be an equilibrium and define

$$e := x - x^*.$$

Definition 3 (Quadratic Lyapunov candidate)**.** *Let*

$$P = \begin{pmatrix} p_E & 0 & 0 \\ 0 & p_N & 0 \\ 0 & 0 & p_\tau \end{pmatrix}, \qquad p_E, p_N, p_\tau > 0.$$

Define

$$V(e) := \frac{1}{2} e^\top P e.$$

Under constant input, the error dynamics satisfy

$$\dot{e} = J^* e + R(e),$$

where

$$\|R(e)\| \le c_R \|e\|^2$$

near the origin.

Let

$$Q := -\big((J^*)^\top P + P J^*\big).$$

Theorem 2 (Local nonlinear Lyapunov stability)**.** *Assume there exists $P \succ 0$ such that*

$$Q \succ 0.$$

Then the equilibrium x^* is locally asymptotically stable for the full nonlinear system.

Proof. Differentiating V along trajectories gives

$$\dot{V}(e) = e^\top P \dot{e} = e^\top P J^* e + e^\top P R(e).$$

Since

$$e^\top P J^* e = -\frac{1}{2} e^\top Q e,$$

we obtain

$$\dot{V}(e) \le -\frac{1}{2}\lambda_{\min}(Q)\|e\|^2 + \|P\| c_R \|e\|^3.$$

Hence $\dot{V}(e) < 0$ for sufficiently small nonzero e, which proves local asymptotic stability. □

Proposition 4 (Local invariance near a stable fusion equilibrium)**.** *Suppose x^* is a locally asymptotically stable equilibrium and*

$$\Phi(x^*) > C_{\text{Nexus}}.$$

Then there exists $\varepsilon > 0$ such that if

$$\|x(0) - x^*\| < \varepsilon,$$

the corresponding trajectory remains in Fusion Mode for all $t \ge 0$.

Proof. The map $\Phi(x) = EN\tau$ is continuous. Since $\Phi(x^*) > C_{\text{Nexus}}$, there exists a neighborhood $\mathcal{N}$ of x^* such that

$$x \in \mathcal{N} \implies \Phi(x) > C_{\text{Nexus}}.$$

Local asymptotic stability implies there exists $\varepsilon > 0$ such that trajectories starting within ε of x^* remain in $\mathcal{N}$ for all $t \ge 0$. □

Remark 1. This result gives persistence of Fusion Mode in a neighborhood of a stable fusion equilibrium, but it is local rather than global.

10 Forward Invariance of Fusion Mode

Whenever $E, N, \tau > 0$, differentiating $\Phi(x) = EN\tau$ along trajectories gives

$$\dot{\Phi} = \dot{E}\, N\tau + E\, \dot{N}\, \tau + EN\, \dot{\tau}.$$

Substituting the state equations yields

$$\begin{aligned}\dot{\Phi} = \Phi\Bigg[&-(\beta_E + \beta_N + \beta_\tau) + \frac{\alpha_E u_E}{E} + \frac{\alpha_N u_N}{N} + \frac{\alpha_\tau u_\tau}{\tau} \\ &+ (\kappa_{EN} + \kappa_{NE})N + (\kappa_{E\tau} + \kappa_{\tau E})\tau + (\kappa_{N\tau} + \kappa_{\tau N})\tau - \gamma_E E - \gamma_N N - \gamma_\tau \tau \Bigg]. \end{aligned} \tag{7}$$

Define

$$\Xi(x,u) := -(\beta_E+\beta_N+\beta_\tau)+\frac{\alpha_E u_E}{E}+\frac{\alpha_N u_N}{N}+\frac{\alpha_\tau u_\tau}{\tau}+(\kappa_{EN}+\kappa_{NE})N+(\kappa_{E\tau}+\kappa_{\tau E})\tau+(\kappa_{N\tau}+\kappa_{\tau N})\tau-\gamma_E E-\gamma_N N-\gamma_\tau \tau.$$

Then

$$\dot{\Phi} = \Phi\, \Xi(x,u).$$

Theorem 3 (Forward invariance of Fusion Mode)**.** *Assume that for every state x satisfying*

$$\Phi(x) = C_{\text{Nexus}}, \qquad E > 0,\ N > 0,\ \tau > 0,$$

and for every admissible control u, one has

$$\Xi(x, u) \geq 0.$$

Then the set

$$\mathcal{F} := \{x \in \mathbb{R}^3_{\geq 0} : \Phi(x) \geq C_{\text{Nexus}}\}$$

is forward invariant.

Proof. Let $x(t)$ be a trajectory with $x(t_0) \in \mathcal{F}$. Suppose it leaves $\mathcal{F}$. By continuity of $x(\cdot)$ and $\Phi(\cdot)$, there exists a first exit time $t_1 \geq t_0$ such that

$$\Phi(x(t_1)) = C_{\text{Nexus}}, \qquad \Phi(x(t)) \geq C_{\text{Nexus}} \text{ for } t \in [t_0, t_1].$$

At such a first exit time one must have $\dot{\Phi}(x(t_1)) \leq 0$. But (7) gives

$$\dot{\Phi}(x(t_1)) = \Phi(x(t_1))\, \Xi(x(t_1), u(t_1)) \geq 0,$$

a contradiction. □

Remark 2. This is a boundary-condition criterion for persistence of Fusion Mode.

11 Optimal Control

Let

$$(z)_+ := \max\{z, 0\}.$$

For a finite horizon $T > 0$, define

$$J(u) = \int_0^T \Big[q_\Phi \big(C_{\text{Nexus}} - \Phi(x(t))\big)_+^2 + q_E E(t)^2 + q_N N(t)^2 + q_\tau \tau(t)^2 + r_E u_E(t)^2 + r_N u_N(t)^2 + r_\tau u_\tau(t)^2 \Big]\, dt.$$

Let

$$L(x, u) := q_\Phi \big(C_{\text{Nexus}} - EN\tau\big)_+^2 + q_E E^2 + q_N N^2 + q_\tau \tau^2 + r_E u_E^2 + r_N u_N^2 + r_\tau u_\tau^2.$$

Let the costate be

$$\lambda(t) = \begin{pmatrix} \lambda_E(t) \\ \lambda_N(t) \\ \lambda_\tau(t) \end{pmatrix}.$$

The Hamiltonian is

$$H(x, u, \lambda) = L(x, u) + \lambda^\top f(x, u).$$

For interior controls,

$$\frac{\partial H}{\partial u_E} = 2 r_E u_E + \alpha_E \lambda_E = 0,$$

$$\frac{\partial H}{\partial u_N} = 2 r_N u_N + \alpha_N \lambda_N = 0,$$

$$\frac{\partial H}{\partial u_\tau} = 2r_\tau u_\tau + \alpha_\tau \lambda_\tau = 0.$$

Hence

$$u_E^\sharp = -\frac{\alpha_E}{2r_E}\lambda_E, \qquad u_N^\sharp = -\frac{\alpha_N}{2r_N}\lambda_N, \qquad u_\tau^\sharp = -\frac{\alpha_\tau}{2r_\tau}\lambda_\tau.$$

Under box constraints

$$0 \le u_i(t) \le u_i^{\max},$$

the optimal control is

$$u_i^*(t) = \Pi_{[0,u_i^{\max}]}\left(-\frac{\alpha_i}{2r_i}\lambda_i(t)\right),$$

for $i \in \{E, N, \tau\}$, where

$$\Pi_{[a,b]}(z) := \min\{b, \max\{a, z\}\}.$$

Theorem 4 (Pontryagin necessary conditions)**.** *Let u^* be an optimal control and $x^*(\cdot)$ its associated trajectory. Then there exists an absolutely continuous costate $\lambda(\cdot)$ such that*

$$\dot{\lambda} = -\frac{\partial H}{\partial x}, \qquad \lambda(T) = 0,$$

and, for almost every $t \in [0, T]$,

$$H(x^*(t), u^*(t), \lambda(t)) = \min_{u \in U} H(x^*(t), u, \lambda(t)),$$

where

$$U = [0, u_E^{\max}] \times [0, u_N^{\max}] \times [0, u_\tau^{\max}].$$

Remark 3. Because the Hamiltonian is convex in the control variables, the minimizing control is obtained by pointwise projection of the unconstrained minimizer.

12 Reachability of Fusion Mode Under Bounded Controls

Assume controls are bounded by

$$0 \le u_i(t) \le u_i^{\max}, \qquad i \in \{E, N, \tau\}.$$

Let

$$\bar{u} := (u_E^{\max}, u_N^{\max}, u_\tau^{\max})^\top.$$

Consider the scalar comparison systems

$$\dot{\underline{E}} = -\beta_E \underline{E} + \alpha_E u_E^{\max} - \gamma_E \underline{E}^2, \tag{8}$$

$$\dot{\underline{N}} = -\beta_N \underline{N} + \alpha_N u_N^{\max} - \gamma_N \underline{N}^2, \tag{9}$$

$$\dot{\underline{\tau}} = -\beta_\tau \underline{\tau} + \alpha_\tau u_\tau^{\max} - \gamma_\tau \underline{\tau}^2. \tag{10}$$

Let $(\underline{E}^*, \underline{N}^*, \underline{\tau}^*)$ denote the positive equilibria of (8)–(10), namely

$$\underline{E}^* = \frac{-\beta_E + \sqrt{\beta_E^2 + 4\gamma_E \alpha_E u_E^{\max}}}{2\gamma_E},$$

$$\underline{N}^* = \frac{-\beta_N + \sqrt{\beta_N^2 + 4\gamma_N \alpha_N u_N^{\max}}}{2\gamma_N},$$

$$\underline{\tau}^* = \frac{-\beta_\tau + \sqrt{\beta_\tau^2 + 4\gamma_\tau \alpha_\tau u_\tau^{\max}}}{2\gamma_\tau}.$$

Theorem 5 (Sufficient reachability condition)**.** *Suppose*

$$\underline{E}^* \underline{N}^* \underline{\tau}^* > C_{\text{Nexus}}.$$

Then there exists an admissible control, namely

$$u(t) \equiv \bar{u},$$

such that Fusion Mode is reachable from any strictly positive initial condition

$$x(0) \in \mathbb{R}^3_{>0}.$$

More precisely, there exists $T > 0$ such that

$$\Phi(x(T)) \geq C_{\text{Nexus}}.$$

Proof. Under the constant maximal control $\bar{u}$, the full dynamics satisfy

$$\dot{E} \geq -\beta_E E + \alpha_E u_E^{\max} - \gamma_E E^2,$$

$$\dot{N} \geq -\beta_N N + \alpha_N u_N^{\max} - \gamma_N N^2,$$

$$\dot{\tau} \geq -\beta_\tau \tau + \alpha_\tau u_\tau^{\max} - \gamma_\tau \tau^2,$$

because all bilinear coupling terms are nonnegative. By scalar comparison,

$$E(t) \geq \underline{E}(t), \qquad N(t) \geq \underline{N}(t), \qquad \tau(t) \geq \underline{\tau}(t)$$

for all $t \geq 0$. Each comparison subsystem converges to its positive equilibrium. Therefore

$$\liminf_{t\to\infty} \Phi(x(t)) \geq \underline{E}^* \underline{N}^* \underline{\tau}^* > C_{\text{Nexus}}.$$

By continuity of $\Phi(x(t))$, there exists a finite $T > 0$ such that $\Phi(x(T)) \geq C_{\text{Nexus}}$. □

13 Finite-State Abstraction

Define

$$\mathcal{M}(x) = \begin{cases} \text{Idle}, & \Phi(x) = 0, \\ \text{Build}, & 0 < \Phi(x) < C_{\text{Nexus}},\ \tau > \tau_{\min},\ E < E_{\max}, \\ \text{Fusion}, & \Phi(x) \geq C_{\text{Nexus}},\ \tau > \tau_{\min},\ E < E_{\max}, \\ \text{Collapse}, & \tau \leq \tau_{\min} \text{ or } E \geq E_{\max}. \end{cases}$$

Set

$$m(t) := \mathcal{M}(x(t)).$$

Theorem 6 (Soundness)**.** *Assume $x(\cdot)$ is a continuous trajectory of the nonlinear state system. Then every transition of $m(t)$ can occur only when the trajectory intersects one of the guard hypersurfaces*

$$\Phi(x) = 0, \qquad \Phi(x) = C_{\text{Nexus}}, \qquad \tau = \tau_{\min}, \qquad E = E_{\max}.$$

Proof. The mode map partitions state space into regions defined by threshold inequalities. Since $x(t)$ and the defining functions are continuous, a mode change can occur only through a boundary crossing. □

Assumption 2 (Transversal guard crossing)**.** *Whenever a trajectory intersects one of the guard hypersurfaces*

$$\Phi(x) = 0, \qquad \Phi(x) = C_{\text{Nexus}}, \qquad \tau = \tau_{\min}, \qquad E = E_{\max},$$

the corresponding scalar guard function has nonzero time derivative at the crossing time.

Theorem 7 (Completeness under transversal crossings)**.** *Under the transversal guard crossing assumption, every continuous guard crossing induces a unique transition in the abstract mode signal $m(t)$. Hence the finite-state abstraction is complete with respect to guard crossings.*

Proof. Let t_c be a crossing time for a scalar guard function $g(x(t))$ satisfying

$$g(x(t_c)) = 0, \qquad \frac{d}{dt} g(x(t_c)) \neq 0.$$

By continuity and nonzero derivative, $g(x(t))$ changes sign across t_c. Since the mode map is defined by the sign structure of the guard inequalities, the mode before t_c and the mode after t_c must differ. This yields a discrete transition. Uniqueness follows from transversality. □

Remark 4. Without transversality, tangential contact with a guard may fail to induce a discrete transition.

14 Global and ISS-Type Estimates

Consider the quadratic storage function

$$V(x) := \frac{1}{2}\left(p_E E^2 + p_N N^2 + p_\tau \tau^2\right), \qquad p_E, p_N, p_\tau > 0.$$

Differentiating along trajectories gives

$$\begin{aligned}
\dot{V} &= p_E E\dot{E} + p_N N\dot{N} + p_\tau \tau\dot{\tau} \\
&= -p_E\beta_E E^2 - p_N\beta_N N^2 - p_\tau\beta_\tau\tau^2 + p_E\alpha_E E u_E + p_N\alpha_N N u_N + p_\tau\alpha_\tau\tau u_\tau \\
&\quad + p_E\kappa_{EN}E^2 N + p_E\kappa_{E\tau}E^2\tau + p_N\kappa_{NE}EN^2 + p_N\kappa_{N\tau}N^2\tau \\
&\quad + p_\tau\kappa_{\tau E}E\tau^2 + p_\tau\kappa_{\tau N}N\tau^2 - p_E\gamma_E E^3 - p_N\gamma_N N^3 - p_\tau\gamma_\tau\tau^3.
\end{aligned} \tag{11}$$

Theorem 8 (Global ultimate boundedness)**.** *Assume controls are essentially bounded:*

$$0 \leq u_i(t) \leq u_i^{\max}.$$

Then every trajectory of the nonlinear system is globally bounded and enters a compact set in finite time and remains there thereafter.

Proof. Using Young's inequality on the mixed control terms, each term such as $p_E\alpha_E E u_E$ is bounded above by a quadratic state term plus a constant depending on $u_E^{\max}$. The positive coupling terms in (11) have total degree three, but the negative cubic saturation terms dominate them for sufficiently large state norm. Therefore $\dot{V}(x) < 0$ outside a sufficiently large compact sublevel set. Standard Lyapunov arguments imply global boundedness and ultimate boundedness. □

Proposition 5 (ISS-type estimate)**.** *There exist class-$\mathcal{K}_\infty$ functions $\underline{\alpha}, \overline{\alpha}, \gamma$ and a class-$\mathcal{KL}$ function β such that solutions satisfy an estimate of the form*

$$\|x(t)\| \leq \beta(\|x(0)\|, t) + \gamma\left(\operatorname*{ess\,sup}_{0\leq s\leq t} \|u(s)\|\right).$$

Proof sketch. The function $V(x)$ is proper on $\mathbb{R}^3_{\geq 0}$, and (11) yields a dissipation inequality of the form

$$\dot{V} \leq -\alpha(\|x\|) + \sigma(\|u\|)$$

for suitable comparison functions α, σ. Standard ISS-Lyapunov theory gives the stated estimate. □

15 Integrated Result

Theorem 9 (Extended integrated fusion theorem)**.** *Consider the controlled nonlinear system with nonnegative bounded controls and nonnegative initial conditions. Then:*

1. *the nonnegative orthant is positively invariant;*
2. *Fusion Mode is characterized exactly by*
$$\Phi(x(t)) \geq C_{\text{Nexus}};$$
3. *any equilibrium $x^* \in \mathbb{R}^3_{>0}$ satisfying*
$$\Phi(x^*) \geq C_{\text{Nexus}}$$
is a steady-state fusion point;
4. *if there exists $P \succ 0$ such that*
$$(J^*)^\top P + PJ^* \prec 0,$$
then x^* is locally asymptotically stable for the nonlinear system;
5. *if moreover $\Phi(x^*) > C_{\text{Nexus}}$, then Fusion Mode is locally invariant near x^*;*
6. *if $\Xi(x, u) \geq 0$ on the boundary $\Phi(x) = C_{\text{Nexus}}$, then the set*
$$\{x : \Phi(x) \geq C_{\text{Nexus}}\}$$
is forward invariant;
7. *if*
$$\underline{E}^* \underline{N}^* \underline{\tau}^* > C_{\text{Nexus}},$$
then Fusion Mode is reachable under admissible bounded control;
8. *the finite-state abstraction is sound with respect to guard crossings;*
9. *under transversal guard crossings, the abstraction is complete with respect to guard crossings;*
10. *every trajectory is globally ultimately bounded, and the system satisfies an ISS-type estimate with respect to bounded inputs.*

16 Appendix: Matching TLA+ Abstraction

The following abstract TLA+ model matches the finite-state part of the paper.

```
------------------------------ MODULE NexusFusion ------------------------------
EXTENDS Naturals, Reals, TLC

CONSTANTS C_Nexus, TauMin, EMax

VARIABLES E, N, Tau, Mode

Phi == E * N * Tau

TypeOK ==
    /\ E \in Real
    /\ N \in Real
    /\ Tau \in Real
    /\ E >= 0.0
    /\ N >= 0.0
    /\ Tau >= 0.0
    /\ Mode \in {"Idle", "Build", "Fusion", "Collapse"}

Init ==
    /\ E = 0.1
    /\ N = 0.1
    /\ Tau = 0.1
    /\ Mode = "Build"

BuildStep ==
    /\ Mode = "Build"
    /\ E' \in Real /\ E' >= E
    /\ N' \in Real /\ N' >= N
    /\ Tau' \in Real /\ Tau' >= Tau
    /\ IF E' * N' * Tau' >= C_Nexus
          THEN Mode' = "Fusion"
          ELSE Mode' = "Build"

FusionStep ==
    /\ Mode = "Fusion"
    /\ E' \in Real /\ E' >= 0.0
    /\ N' \in Real /\ N' >= 0.0
    /\ Tau' \in Real /\ Tau' >= 0.0
    /\ IF Tau' <= TauMin \/ E' >= EMax
          THEN Mode' = "Collapse"
          ELSE /\ Mode' = "Fusion"
               /\ E' * N' * Tau' >= C_Nexus

CollapseRecovery ==
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```
    /\ Mode = "Collapse"
    /\ E' \in Real /\ E' >= 0.0
    /\ N' \in Real /\ N' >= 0.0
    /\ Tau' \in Real /\ Tau' >= 0.0
    /\ IF Tau' > TauMin /\ E' < EMax
          THEN Mode' = "Build"
          ELSE Mode' = "Collapse"

Next ==
    BuildStep \/ FusionStep \/ CollapseRecovery

Spec == Init /\ [][Next]_<<E, N, Tau, Mode>>
==============================================================================
```

17 Conclusion

The Nexus Triple Product Fusion Model is formalized as a nonlinear control system with threshold-defined operating modes, equilibrium structure, local nonlinear stability, a control-theoretic formulation, reachability and invariance results, and a verification-oriented abstraction. This document brings the mathematical, diagrammatic, and verification-oriented components into one file.